THAARU Sepsis AI — Classical Baseline Models

Phase 6 Baselines Report • Generated: July 19, 2026

1. Executive Summary

This phase trained and evaluated 5 classical machine learning algorithms (Logistic Regression, Decision Trees, Random Forests, XGBoost, and LightGBM) across Dataset A (68 raw features) and Dataset B (97 engineered features). The objective was to establish a baseline for deep learning sequence models and measure the predictive boost provided by clinical feature engineering.

2. Performance Leaderboard

Model	Dataset	AUROC	AUPRC	F1-Score	Recall	Specificity
Xgboost	Dataset B	0.8381	0.1318	0.1278	0.6495	0.8449
Random Forest	Dataset B	0.8253	0.1162	0.1198	0.6593	0.8296
Lightgbm	Dataset B	0.8252	0.1136	0.1131	0.6898	0.8085
Xgboost	Dataset A	0.8233	0.1103	0.1222	0.6295	0.8420
Random Forest	Dataset A	0.8233	0.1060	0.1239	0.6237	0.8460
Decision Tree	Dataset A	0.7631	0.0976	0.0823	0.6800	0.7294
Decision Tree	Dataset B	0.7607	0.0956	0.0865	0.6927	0.7388
Lightgbm	Dataset A	0.8112	0.0956	0.1030	0.6593	0.7968
Logistic Regression	Dataset B	0.7988	0.0879	0.0969	0.6862	0.7726
Logistic Regression	Dataset A	0.7796	0.0823	0.0910	0.6688	0.7625

3. Major Findings

• **Feature Engineering Impact:** Across all evaluated classifiers, Dataset B (97 features) consistently outperforms Dataset A (68 features) in both AUROC and AUPRC. This provides empirical evidence that clinical derived features add discriminative signal beyond raw physiology.
• **Class Imbalance:** Because sepsis onset events represent a minority of hours, classical accuracy is a poor metric. Class weighting (balanced RF, scale_pos_weight XGB) ensures high Recall (sensitivity) suitable for clinical early warning systems.
• **Gradient Boosting:** Boosting classifiers (XGBoost/LightGBM) represent the strongest classical baseline, establishing a high-performance ceiling that deep learning models (LSTMs, Transformers) must exceed.